EPIPHYTE DIVERSITY ON TREE TRUNKS

Updated: October 26, 2025



Figure 1. Lichens *Usnea* (top) and *Evernia* (bottom).

Pre-Lab Resources on Canvas:

(1) Epiphyte field guide
(2) Epiphyte ID video
(3) Epiphyte ID practice quiz

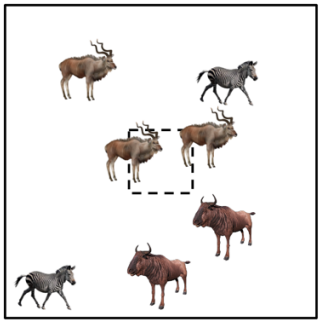
Background Information & Key Concepts

This lab investigates epiphytes (things that grow on plants; Fig. 1) on trees around UBC campus.

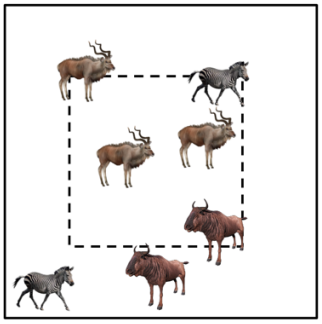
<u>Species-area relationships</u>

Species richness increases with area. Three processes drive this pattern:

(1) Sampling effect: As area increases, more individuals are included in the sample, increasing the chance that you will capture new species based on probability alone.

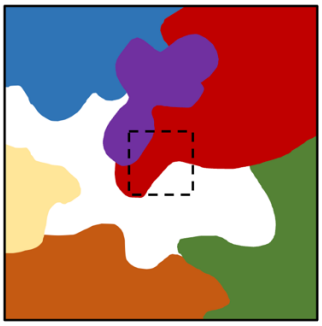


1 species captured

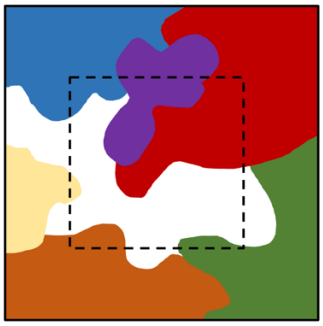


3 species captured

(2) Habitat effect: As area increases, more habitats are included in the sample. This increases the number of species captured, assuming species composition varies between habitats. For epiphytes, one tree can contain many different micro-habitats (crevices within bark, vertical position on tree, etc.).

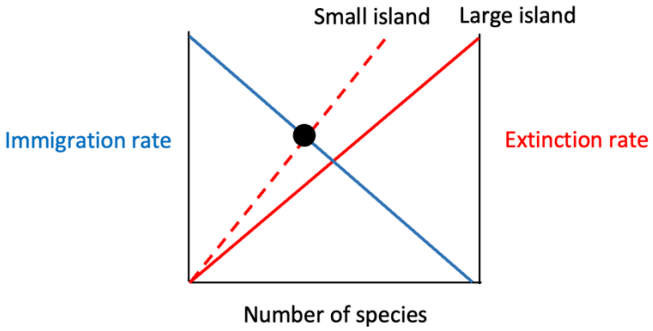


3 habitats captured



7 habitats captured

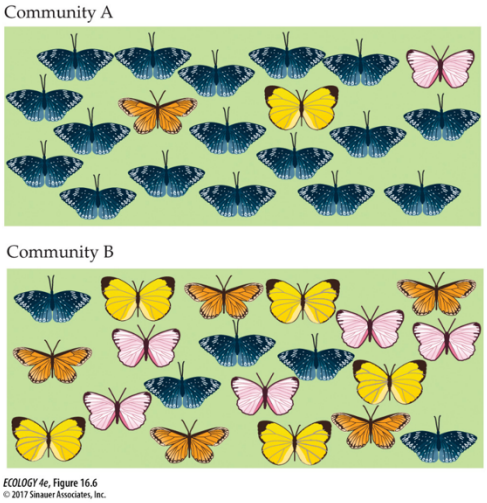
(3) Island biogeography: As area increases, extinction rates decrease, increasing the number of species.

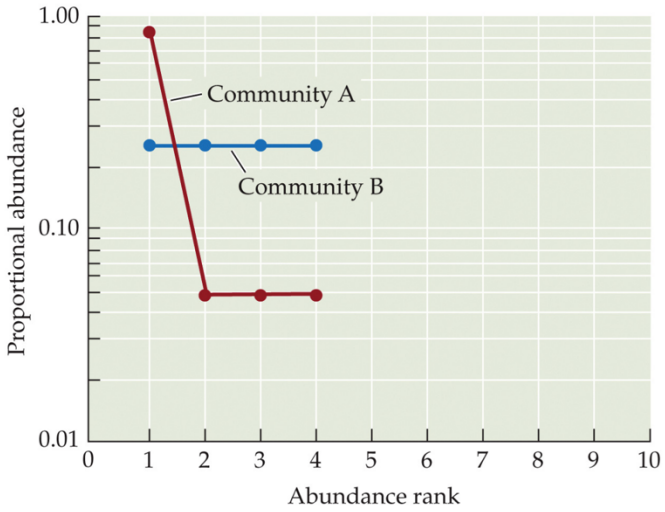


Consider how sampling effect, habitat effect, and IBT could apply to epiphytes growing in trees. In other words, how does the size of a tree affect the number of epiphyte species it supports?

Rank-abundance curves

Rank-abundance curves, in which species within a community are ordered from most- to least-abundant, help us visualize species richness and abundances.





Materials

For every working group:

- Clipboard
- Datasheet
- Epiphyte guide
- Quadrat sheet
- Meter tape
- Pushpins
- Hand lenses

For the TA:

- Clipboard
- Compass
- Epiphyte guide
- Attendance sheets, pencils, extra datasheets, maps
- First aid kit

Field Methods



Figure 2. Pinning a transparent quadrat to a tree.

Meet your TA at the gathering location (will be posted on Canvas; same as PSP lab), then walk to your sampling area on campus. Your group will be assigned a specific set of three trees for sampling, labelled with metal tags.

Step 1. For each tree, record the tag number and tree size. For tree size, measure the circumference (to the nearest whole cm) with the tape at 150cm above the ground. After the lab, circumference must be converted to DBH (see below).
Step 2. Using a compass on your phone, determine which side faces north, south, east and west. If you do not have a compass, the TA can use theirs to help you. On each side, you will sample at three heights off the ground: 0, 75 and 150cm. In other words, you will sample 12 times per tree.
Step 3. For each sample, gently pin the transparent quadrat (3x3 cells) onto the bark in portrait orientation (Fig. 2), with the lower edge at the specific height from step 2. Identify all bryophyte and lichen species present within the quadrat and record the species codes (see list of codes on datasheet). For each species, count how many of the 9 cells it is present in. The count data will therefore be a number between 0 and 9.
Step 4. Repeat for 11 more samples, then repeat all steps for 2 more trees.

After the Lab

Share your datasheet among group members (take pictures and email to one another). One group member uploads the datasheet to our website (www.zoology.ubc.ca/EcologyLabs/).

IMPORTANT: Before uploading, follow these steps:

(1) Convert tree circumference to diameter at breast height (DBH) by dividing circumference by π
(2) Average each epiphyte column to a single value per tree (include the zeroes). This gives us an average abundance of each epiphyte per tree.
(3) Upload epiphyte data with enough decimal places that small, non-zero values don't just appear as zeroes (e.g. 0.003, not 0.00).

If you skip one of these steps, the data will be wrong!

Ensure your group member has uploaded the data before the deadline. Once Rachel has cleaned up the data, you will be able to download the full class dataset and proceed to the assignment on the next page.

Pre-Analysis

This lab requires you to use some techniques learned in the Excel Activity; revisit this activity if you need a refresher. As always, visually inspect the data and review the metadata before attempting your analysis.

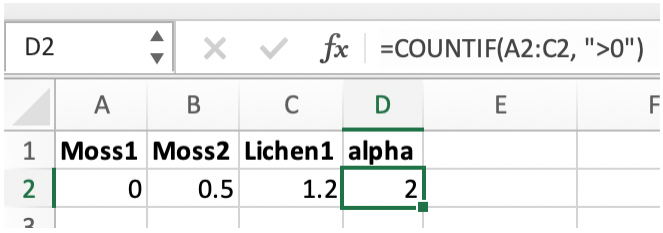
Optional: You can use the "Freeze Panes" tool to ensure your columns headers are visible as you scroll through your data. Click the leftmost cell immediately below your column header, then click Window → Freeze Panes. You can always click Window → Unfreeze Panes if you freeze from the wrong cell.

Data Analysis Task 1: Species-area relationships

First, we will calculate species richness of epiphytes (bryophytes and lichens) on every individual tree. Then, we will determine whether there is a relationship between tree size (diameter) and epiphyte species richness.

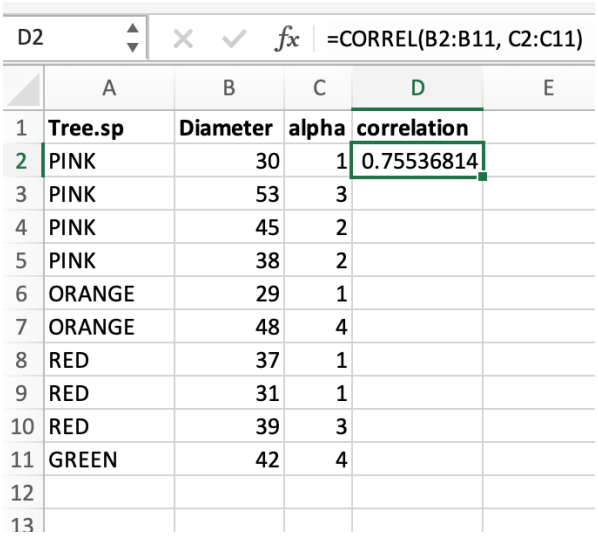
(1) Calculate richness of epiphytes on individual trees by counting the number of non-zero epiphyte abundances for a given tree. You can do this using a COUNTIF function.

In the example below, cells A2-C2 each give the average abundance of a different epiphyte species on one individual tree, so a COUNTIF function will return the value "2" in cell D2; i.e., 2 epiphyte species occurred on that particular tree. Make sure you don't accidentally include Diameter.cm (tree diameter) in your count!



(2) Using a CORREL function, calculate the Pearson's correlation coefficient (*r*) between tree size (measured as diameter at breast height) and species richness per tree (calculated above). Return to the Excel Activity if you need a refresher on correlation coefficients. Note that you will get the same answer if you switch the order of the variables in the formula.

Select the full range of diameter values and epiphyte richness values, as in the below example:

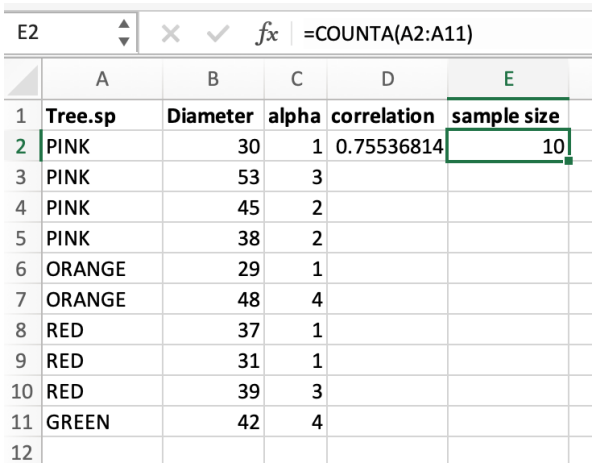


D2 =CORREL(B2:B11, C2:C11)

	A	B	C	D	E
1	**Tree.sp**	**Diameter**	**alpha**	**correlation**	
2	PINK	30	1	0.75536814	
3	PINK	53	3		
4	PINK	45	2		
5	PINK	38	2		
6	ORANGE	29	1		
7	ORANGE	48	4		
8	RED	37	1		
9	RED	31	1		
10	RED	39	3		
11	GREEN	42	4		
12					
13					

(3) Calculate the sample size for this correlation as the total number of trees we surveyed. There are a few different ways to do this, but one way is to use a COUNTA function, which counts non-numeric data. You don't need to specify a criterion, just select the range of data containing your tree species (or tree number). Make sure to exclude the header row!

See below for an example:



E2 =COUNTA(A2:A11)

	A	B	C	D	E
1	**Tree.sp**	**Diameter**	**alpha**	**correlation**	**sample size**
2	PINK	30	1	0.75536814	10
3	PINK	53	3		
4	PINK	45	2		
5	PINK	38	2		
6	ORANGE	29	1		
7	ORANGE	48	4		
8	RED	37	1		
9	RED	31	1		
10	RED	39	3		
11	GREEN	42	4		
12					

(4) At this point, we know the direction (positive vs negative), strength (closer to 0 vs closer to 1), and sample size of our correlation, but we don't know if the relationship is significant. In other words, is our correlation truly different from a correlation of zero, or could it have been obtained purely due to random chance? We can find out by calculating a P-value.

Enter your correlation coefficient and sample size into the text boxes here:
https://www.danielsoper.com/statcalc/calculator.aspx?id=44

Record the two-tailed P-value only. If the P-value is <0.05, the trend is significant and likely did not occur due to chance. If the p-value is >0.05, it is not significant, meaning that any apparent correlation you see in the data is statistically indistinguishable from no correlation at all.

(5) Prepare a scatterplot showcasing this relationship, with tree DBH on the x-axis and species richness on the y-axis (Figure 1). Add a trendline and display the R-squared value. Include a caption and report Pearson's *r* and the P-value in the caption to 2 decimal places (see the lab manual for an example of how to do this). For any values < 0.01, use scientific notation to 2 decimal places. Use the formatting tips from your Excel Activity feedback, PSP Lab feedback, and the lab manual.
Tip: Fig. 1 will just display floating points with no connecting lines, unlike Fig. 2 (see below).

Data Analysis Task 2: Rank-abundance curves

In Task 1, we calculated species richness of epiphytes on each tree from our abundance data (measured as average number of quadrat cells an epiphyte was found in across 12 samples – see field methodology). Now, we will use these same data to create a rank-abundance curve for epiphytes. Because each tree hosts its own mini-community of epiphytes, we could theoretically create a rank-abundance curve for every tree in the dataset. However, this would take a lot of time (and interpreting over 100 curves would be rather difficult)!

Instead, we will average the abundance of each epiphyte species across all trees to give insight into "average" community composition.

(6) For each epiphyte species, use the AVERAGE function to calculate an average abundance across all trees. It's recommended that you enter these calculations at the bottom of your spreadsheet so that you have room to write the formula once and drag to fill in values for all species. Return to the Excel Activity if you can't remember how to drag.

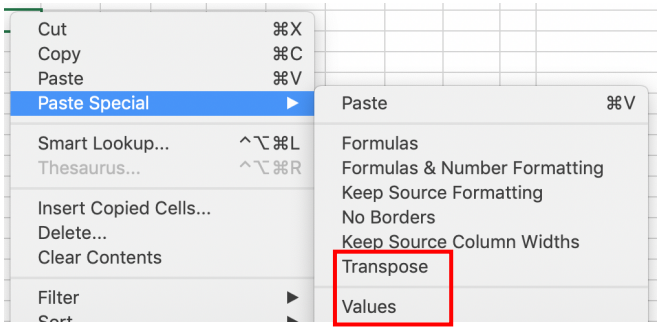
(7) Now, we will create a rank-abundance curve. Our graph will show mean abundance (calculated in step 6, then ordered from most to least abundant) on the y-axis and species rank on the x-axis.

To get the species in the right order, you'll need to copy, sort, and rank your abundances (steps 7-9).

First, make a numeric copy (i.e., no formulas) of your average epiphyte abundances. Select your row of average abundances, copy it, right-click in the destination field (e.g., right below your original averages), select "Paste Special", then choose "Values" (see below). **If you do not complete this step, the next step will not work.**

(8) Unfortunately, Excel can't plot your data correctly if it's horizontally oriented. Select and copy the numeric data, then right-click in a new destination field (I recommend under "Mean abundance" at

the top of your spreadsheet), select “Paste Special”, and select “Transpose” (see below). This will convert your horizontally-oriented data to vertically-oriented data.



(9) Once you have your abundances in a vertical orientation, select **just those cells**, and use Data > Sort to sort based on the abundance column (under “Column” pulldown), largest to smallest (under “Order” pulldown). Note that there is a little box in the far right that says “my data has headers” which you should either click or unclick depending on whether you selected the header row. If you get a warning about data outside current selection not being sorted, choose to continue with the current selection.
IMPORTANT: Before you execute the sort, you need to be 100% certain you have only selected the range of data you want to manipulate; otherwise, your previous analyses could get jumbled! If you want to be extra-careful, you can copy your data into a new sheet before sorting.

(10) Your mean abundances should now be sorted from largest to smallest abundance. Fill in a species rank to the left of the abundance column. The rank column will be a numbered series 1-S, S being the total number of epiphyte species we looked at.

(11) Finally, create a scatterplot with rank on the x-axis and mean abundance on the y-axis (Figure 2). Add a line connecting the points by opening the Format Data tab, clicking the Fill icon, and checking off Solid Line. The plot should resemble the curves we looked at in class (you can also search online for other examples). Make sure you add an informative caption and use the formatting tips from your Excel Activity feedback, PSP Lab feedback, and the lab manual. Your sample size is the same as the sample size calculated in Task 1. Do not include a trendline or R-squared.

See next page for lab assignment instructions.

Lab Assignment [18 MARKS TOTAL]

Due: October 31, 5:00 PM (buffer to 11:59 PM)

The report should occupy no more than 2 pages. Page 1 is for Figures 1 and 2 and their captions; Page 2 your written answers with margins no less than 1 inch (try to keep it to about half a page of writing). See lab manual for additional details on formatting and general guidelines.

Please also submit your Excel spreadsheet.

Figures and Captions [8 MARKS TOTAL]

Questions [9 MARKS TOTAL]

1. In the introduction, we discussed how sampling effect, habitat effect, and island biogeography theory can drive relationships between area and species richness. Given the way we sampled, do you think any of these processes could be driving the hypothetical relationship between epiphyte species richness and tree size in our lab? Explain in detail. [4 marks]
Hint: this is a conceptual question, so you don't need to consider your results in your answer.

2. Describe the relationship between epiphyte species richness and tree size from Fig. 1 and your correlation analysis. Are the results expected or unexpected? Explain. [3 marks]

3. In general, how do patterns in rank-abundance curves relate to species evenness? Given this, does Fig. 2 suggest an even or uneven community? [2 marks]

Clarity of Writing [1 MARK TOTAL]